

David Santiago Puentes Cárdenas

Full-Stack Software Developer · B2B SaaS · IoT & Applied AI

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PROFILE

Full-stack developer and electronic-engineering student with more than two years of professional experience building, integrating, and supporting products used by real companies. At PRLCOL I own the full delivery cycle — requirements, design, development, SQL migrations, deployment, and support — for 7 enterprise customers of a multi-instance B2B SaaS platform, with continuous deployment to production. My electronics background lets me work from firmware and binary protocols (LoRa, BLE, Teltonika) up to the web dashboard. I integrate LLMs into products with cost controls and use AI-first tooling (Claude Code, opencode) with human review before every deployment.

EXPERIENCE

Full-Stack Software Developer & Data Analyst · PRLCOL

Aug 2025 — present · Tocancipá, Colombia

- Own the full cycle for 7 enterprise customers (food manufacturing, energy, oil and gas, engineering) of a multi-instance B2B SaaS platform for operations and workplace-safety management: each customer has its own subdomain, database, branding, and deployment.
- Designed and shipped end to end a high-risk work-permit system: Planning → Approval → Execution → Closure flow, digital signatures drawn on the actual PDF (pdf.js + canvas), remote signing through single-use tokenized links with email OTP, two-level drafts, and a signature lock; in live operation at two industrial plants (~340 commits).
- Integrated an LLM assistant (Gemini API) over operational data: contextual chat, predictive reports, per-module and per-month context, a monthly token cap per company, an attempt log, a support panel, and prompt-injection defenses.
- Built an end-to-end GPS telemetry system deployed to 4 customers: a custom TCP server in PHP decoding the Teltonika Codec 8/8E binary protocol, Traccar-compatible HTTP ingestion from an Android app, OSRM map-matching, Leaflet geofences, reverse geocoding, alerts, consent records, and Excel/PDF reports (~55,000 lines).
- Built from scratch a public QR self-registration flow for visitors and contractors (digital badge, validity by date and site, gate verification) on a layered architecture of my own: thin controllers, services, repositories with prepared statements, and authentication, CSRF, and rate-limiting middleware.
- Delivered bulk Excel imports with validation and idempotent upserts, dynamic per-module and per-action permissions, branded Excel/PDF generation (PhpSpreadsheet, Dompdf), and HTML transactional email on every state transition; datasets of more than 1.5 million records.
- Hardened authentication across 5 production instances (bcrypt unification, session-collision fix) and applied prepared statements, CSRF protection, rate limiting, upload sanitization, and credential-incident handling platform-wide.
- Led the dashboard redesign and design system replicated across 4 customers, and ported complete modules between instances without breaking production.
- Shipped 987 production commits across 8 repositories in 8 weeks (Jun–Jul 2026) and 47 versioned, zero-downtime SQL migrations, using a staging branch per customer, granular commits tied to kanban cards, customer-approved batch PRs, webhook deployment, and git revert as rollback.

Research & Development Developer · Universidad ECCI / CEINTECCI

Jun 2024 — Jun 2025 · Bogotá, Colombia

- Co-developed GenACT, an executable Python library for stylometric analysis of Greek lyric texts using genetic algorithms. **Patent granted** for genetic algorithms applied to stylometric analysis.
- Built reproducible extraction, cleaning, normalization, and analysis pipelines over roughly 2 million syllables of research datasets.
- Developed and maintained the center's science-outreach site with Astro, TypeScript, React, and Tailwind CSS.
- Supported methodology, results analysis, academic documentation, and research-derived products.

SELECTED PROJECTS

SmartSense Monitoring — industrial thermal monitoring (thesis, two-author team)

Three-wire PT100 + MAX31865 nodes on XIAO ESP32-C6 with LoRa 433 MHz, a gateway with touch display and local dashboard, and a neutral Docker/Flask/SQLite replica. 3 nodes, 51,338 samples, up to 32 days of records, estimated PDR 95.3–96.6% in active periods, and a link demonstrated at 688 m through obstacles (lower bound). Open hardware

(CERN-OHL-S v2 / MIT / CC-BY 4.0); HardwareX article in preparation. Personal contribution: firmware, analysis scripts, replica, documentation, and validation.

Sensor Dashboard — public IoT system (MIT)

MicroPython nodes with LoRa (sx127x) → gateway → Flask API → PostgreSQL (Neon) → Next.js + TypeScript + Tailwind dashboard on Vercel, built alone. Public evidence of React, Next.js, TypeScript, and PostgreSQL work.

Project H — room and asset security

BLE nodes with a custom binary protocol (opcodes, ACK, timeouts), an ESP32 master with a physical alarm and an embedded async web server, a Firebase Auth web app, and Cloud Functions + Cloud Messaging push notifications for a public hospital. Prototype stage.

TECHNICAL SKILLS

Backend: PHP 7.1–8.x (mysqli/PDO, transactions), Python (Flask, pandas/NumPy, SciPy), Node.js (Firebase Cloud Functions), REST APIs, TCP sockets and binary protocols, CLI daemons, layered MVC/SOLID without frameworks

Frontend: JavaScript ES6, jQuery, Chart.js, DataTables, Leaflet, pdf.js, jSignature, Service Workers, Web Push (VAPID); TypeScript, React, Next.js, Astro, Tailwind CSS (public projects)

Databases: MySQL/MariaDB, PostgreSQL, SQLite; relational modeling, indexes, versioned and idempotent migrations, data cleaning and validation

Electronics & IoT: ESP32 (C6, S3), C/C++ (Arduino, ESP-IDF), MicroPython, LoRa, BLE, PT100/MAX31865, deep sleep and battery life, KiCad

Integrations: Gemini API, Firebase (Auth, Firestore, Messaging), ZeptoMail/SendGrid/PHPMailer, OSRM, Traccar/Teltonika, SUIN API, QR generation, Dompdf, PhpSpreadsheet

Delivery & operations: Git/GitHub (staging → main, batch PRs, revert as rollback), webhook-driven continuous deployment, Docker, Linux, Composer, Laragon; Terraform certification (strengthening)

AI-first workflow: Claude Code, opencode; specs before code, plans, subagents, skills, MCP, token budgets, adversarial diff review, and human validation before production

Quality & security: root-cause debugging, exploratory testing in staging, prepared statements, CSRF, rate limiting, bcrypt, session control, credential handling

EDUCATION & CERTIFICATIONS

Electronic Engineering — Universidad ECCI, Bogotá (in progress) · Technologist in Industrial Electronics — Universidad ECCI (2023) · English B2 — Centro de Lenguas, Universidad ECCI (2025)

AI Application Development · React.js with TypeScript · TypeScript · Terraform · Cybersecurity and Privacy for Businesses — Platzi (2026)

Scientific Computing with Python · Responsive Web Design — freeCodeCamp (2025)